

Shivam Murawala

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EDUCATION

Woburn Collegiate Institute

Ontario Secondary School Diploma (Expected 2027), Grade 12

Toronto, ON

Sep. 2023 – Jun. 2027

- **Activities:** Computer Science Club, Math Team, DECA, Robotics, Cross Country Team

EXPERIENCE

Mobile Infrastructure Developer

Nokia

Ottawa, ON

Jul. 2026 – Aug. 2026

- **Resolved 12 blocker bugs** in the Java codebase of Nokia Policy Controller (NPC)
- Onboarded 10+ engineers onto a **centralized machine learning workflow** and configured 2 virtual machines, reducing manual environment setup time by 20%.
- Patched external testing packages, including Rammbock, **improving codebase security** and network reliability.

Founder and President

WCI Computer Science Club

Toronto, ON

Sep. 2025 – Present

- Founded and led a **40+ member club**, driving active participation and successful completion of group projects through weekly web development and programming sessions.
- Partnered with **UTSC Professor Francisco Estrada (Paco)** to host an LLM and Transformer workshop.
- Led the planning and execution of a 2-day school-wide hackathon with **15+ teams**.

Independent Researcher

Toronto Science Fair

Toronto, ON

Sep. 2023 - Mar. 2024

- **Conducted 400+ automated simulations** comparing chess engines, including Stockfish and Leela Chess Zero.
- Analyzed differences between **Minimax-based search** and **MCTS/neural-network** architectures in gameplay.
- Concluded neural-network engines favor long-term position, while Minimax engines excel at tactics.

PROJECTS

Conway's Game of Life Simulator [GitHub] | C++, Raylib

- Built a **high-performance implementation** of John Conway's Game of Life using **C++ and Raylib**
- Implemented real-time controls to start, stop, and modify the simulation state dynamically
- Added support for random grid generation, custom initial states, adjustable simulation speed, and cell colors

ShivamOS, 3D Portfolio [GitHub] | React.js, Three.js, JavaScript

- Built an **interactive 3D portfolio**, featuring a computer floating in space
- Implemented a camera zoom transition into a custom, fully interactive **terminal-based OS**
- Added sound design with **Howler** to enhance immersion during scene transitions

AWARDS & ACHIEVEMENTS

- **Lloyd Auckland Math Workshop @ University of Waterloo** — One of 95 students invited across Canada
- **Canadian Computing Competition (CCC)** — Top 10% and School Champion
- **AIME Qualifier: Top 10% on AMC** — Scored 114/150 on AMC 12B
- **DECA International Career Development Conference** — Placed 3rd globally out of 1,000+ competitors
- **FIRST Robotics Competition World Championship** — Programming (Java) and Impact Team, Team 188
- **CEMC Euclid Contest** — Distinction, Top 10% and, School Champion
- **CEMC Hypatia** — Distinction, Group IV Placement, School Champion
- **Toronto Science Fair** — Silver Medal Recipient

TECHNICAL SKILLS

Languages: C++, Python, Java, JavaScript/TypeScript, HTML5, CSS, SQL

Frameworks & Libraries: WPILib, React.js, Matplotlib, Pandas, Scikit-Learn, Raylib, JUnit5, Three.js, Spring

Tools: Git, GitHub, VS Code, Gradle, Figma, Gerrit, Jira, Docker, WSL, Maven